

REECE DICKINSON

Master Curriculum Vitae & Comprehensive Technical Portfolio

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PROFESSIONAL PROFILE

Highly analytical and results-oriented Computer Science student maintaining a 3.94 GPA as a Presidential Scholar at Cal Poly Humboldt. Possesses a unique dual competency: a robust, low-level technical understanding of UNIX/Linux systems internals, C/C++ architecture, POSIX multiprocessing, and containerized infrastructure, combined with modern, high-level capabilities in full-stack Generative AI system engineering, Retrieval-Augmented Generation (RAG) evaluation harnesses, and local GPU diffusion pipeline optimization (ComfyUI / PyTorch). Proven track record of high-stakes technical leadership, cross-functional problem-solving, and communicating complex technical paradigms to diverse, non-technical stakeholders. Fully bilingual in English and Spanish, equipped to serve as a versatile systems engineer, AI engineer, and technical leader.

EDUCATION & ACADEMIC CREDENTIALS

Cal Poly Humboldt — Arcata, CA

Expected May 2028

Bachelor of Science in Computer Science

- Academic Honors:** Cumulative GPA: 3.94 / 4.00 | Presidential Scholar (Top Academic Honor, 3.95+ Threshold) | Dean's List (Every Academic Term).
- University Leadership:** President, Spanish National Honor Society; Webmaster, Cal Poly Humboldt Aviation Club.
- Comprehensive Coursework History:** Systems Programming (CS 375), Computer Architecture (CS 243), Web-Apps Using Databases (CS 328), Database Design (CS 325), Data Structures (CS 211), Computer Science Foundations 1 & 2 (CS 111, CS 112), Intro to Data Science (DATA 111), Introductory Biostatistics (STAT 109), Discrete Mathematics (MATH 253), Calculus I (MATH 109), Algorithms (CS 312), Operating Systems (CS 374), Telecommunications & Networks (CS 346), Software Engineering (CS 458), Computer Security (CS 449), Computational Models (CS 461), Machine Learning for Data Science (DATA 322).

Arcata High School — Arcata, CA

Aug 2020 – Jun 2024

High School Diploma

- Academic Excellence:** Cumulative High School GPA: 4.10 (Overall 4-year combined GPA: 4.02). Received Academic Office & Principal's Awards.
- Specialized Training:** Completed AP Computer Programming, 4 years of Spanish language, and selected for the Arcata Arts Institute (AAI).

COMPREHENSIVE TECHNICAL DIRECTORY

- Programming Languages:** C, C++, Python (FastAPI, PyTorch, Pandas, NumPy, Scikit-Learn), Java, SQL (PostgreSQL, PGVector, Oracle SQL), JavaScript, TypeScript, Node.js, R, HTML5, CSS3, PHP, Lua, Linux Shell Scripting.
- Systems & Infrastructure:** UNIX/Linux Systems Admin, POSIX Multiprocessing APIs ('fork', 'exec', 'wait'), Low-Level I/O Redirections ('<', '>'), Signal Handling ('SIGINT', 'SIGTSTP'), Valgrind Memory Leak Profiling, Pointer Arithmetic, Computer Architecture, Docker, Kubernetes/Containers, Terraform (Infrastructure-as-Code).
- AI, ML & Vector Engineering:** Retrieval-Augmented Generation (RAG) Systems, Automated RAG Evaluation Harnesses ('rag_eval_harness.py', RAG Triad Benchmarks), Vector Embeddings (PGVector), Page-Level Citation Mapping ('Trust Contract'), Local Diffusion Model Pipelines (Wan 2.2 14B MoE, LTX-2.3 22B, Chroma1-HD), ComfyUI Node-Graph

Optimization, FP8 Quantization, VRAM/GPU Optimization, Real-Time AI StreamDiffusion, Multi-Sensor Depth Ingestion (Kinect), Audio-Video Sync Pipelines (Wan 2.2 + MMAudio), RVC / Audio Voice Synthesis (Applio, Google Colab).

- **Developer Tools & Middleware:** FastAPI, React.js, Node.js, Supabase, Alembic, PyTorch, ComfyUI, TouchDesigner, Git/GitHub, Railway, Oracle SQL Developer, CUDA, Valgrind, RESTful APIs, JSON/YAML Schemas.
- **Certifications & Languages:** Anthropic AI Fluency: Framework & Foundations, Anthropic Claude 101, National Honor Society | Fully Bilingual (Native/Fluent English, High-Proficiency Conversational Spanish).

TECHNICAL PROJECTS & SYSTEMS PORTFOLIO

Full-Stack GenAI Financial Copilot & RAG Evaluation Suite

Jun 2026 – Present

Technologies: Python (FastAPI), React, Supabase, PGVector, Alembic, OpenAI APIs

- **Cloud-Native Architecture & Embeddings:** Architected an end-to-end research platform parsing multi-page 10-K/10-Q SEC financial filings (Apple, NVIDIA, Microsoft, Amazon, Google) with real-time response streaming and PostgreSQL vector embeddings via Alembic schema migrations.
- **Trust Contract & Citation Engine:** Engineered a page-level citation mapping engine anchoring AI outputs to exact document coordinates to eliminate hallucinations, providing clickable source-grounded verification inside the user interface.
- **Automated RAG Evaluation Harness:** Built an automated RAG evaluation harness (``rag_eval_harness.py``) measuring Faithfulness, Context Precision, Context Recall, Answer Relevancy, and Citation Accuracy across 25 financial Q&A benchmarks, achieving a 92.0% pass rate.
- **Beta User Testing & Bug Resolution:** Coordinated hands-on testing with 10 active beta user testers to uncover edge-case bugs, optimize prompt responses, and systematically refine query reliability under real-world usage.

Custom POSIX C Command Shell Interpreter

Fall 2026 – Present

Technologies: C, POSIX APIs, Linux Systems Internals, Valgrind, System Calls

- **Low-Level Multiprocessing & Lifecycle Control:** Engineered a POSIX-compliant C shell managing parent-child process lifecycles (``fork``, ``exec``, ``wait``) and executing concurrent foreground/background tasks with low-level execution control.
- **File Descriptor Redirection & Memory Profiling:** Programmed custom file descriptor manipulation routines for stream redirections (``<``, ``>``) and implemented signal handlers (``SIGINT``, ``SIGTSTP``), maintaining strict memory management with zero Valgrind leaks and zombie process elimination.

Local Generative Media & GPU Workstation Pipeline

Summer 2026 – Present

Technologies: PyTorch, ComfyUI, RTX 4090 Workstation, FP8 Quantization, Node-Graph Engineering

- **GPU Memory Tuning & Source Debugging:** Deployed open-weight image/video diffusion models (Wan 2.2 14B MoE, LTX-2.3 22B) on an RTX 4090 GPU, optimizing FP8 quantization and VRAM efficiency while patching cascading Python dependency breaks (kornia API, huggingface_hub, BigVGAN).
- **Audio-Video Sync & Race Condition Fixes:** Engineered an audio-synchronized video pipeline (Wan 2.2 + MMAudio) by fixing frame-rate truncation bugs and resolving file-handoff race conditions using pure in-memory execution node connections.

Learning Engine — Hackathon for Social Good

March 2026 – Present

Technologies: React, Node.js, JSON, Agile Software Engineering

- **No-Code Content Schema & Onboarding Runbooks:** Co-developed a full-stack study tool letting instructors generate dynamic quizzes using no-code JSON schemas, authoring setup guides for non-technical professors.
- **Database Deadlock Resolution:** Identified, isolated, and resolved a critical database deadlock and performance bottleneck during the final sprint of live development.

Relational Database & Transportation Tracking System

Fall 2025

Technologies: SQL, Oracle SQL Developer, Relational Data Modeling (3NF)

- **Relational Schema Architecture:** Served as database architect for a 3-person team, designing an ER diagram with 25+ entities normalized into 3NF across 20+ tables with complex SQL queries and transaction boundaries.

Probabilistic Duel Simulators & Dice Engines

Spring 2025

Technologies: C++, Object-Oriented Design, Monte Carlo Methods

- **OOP & Monte Carlo Statistical Engine:** Programmed a Yahtzee engine and probabilistic duel simulator in C++ using abstract classes, encapsulation, and operator overloading, analyzing millions of Monte Carlo random-walk runs.

LeetCode Algorithmic Mastery & Linux Homelab

Summer 2026 – Ongoing

Technologies: Data Structures, Algorithms, Linux Administration, Networking

- **Algorithmic Fluency & Homelab Admin:** Completed all 75 Grind 75 problems. Self-hosts multi-user Linux game servers, managing NAT, port-forwarding, and bandwidth allocation.

PROFESSIONAL EXPERIENCE

Cal Poly Humboldt Library — Library Scholar Intern (Terraform AI AR Sandbox)

Feb 2026 – May 2026

Arcata, CA | Sponsored Collaborative Research Project

- **Real-Time Sensor & AI Pipeline Re-Architecture:** Rebuilt a physical AR topology sandbox from scratch, replacing a legacy C++ system by routing Kinect depth camera inputs into a real-time AI StreamDiffusion pipeline (TouchDesigner, local Python server) running at 60+ FPS.
- **Systems Troubleshooting & MIT Open-Source Release:** Diagnosed network hops, driver incompatibilities, and frame-rate bottlenecks alongside IT teams; authored technical onboarding runbooks and published the complete codebase under an open-source MIT License.
- **Cross-Functional Technical Communication:** Translated complex diffusion architecture setbacks into clear briefs for non-technical university project managers and department heads.

Ramy Dickinson Construction — Contractor Assistant / Apprentice

Summer 2023 – Present

Arcata, CA | Small Business Construction Operations

- **Autonomous Job-Site Operations & Logistics:** Manage job-site logistics, equipment operations, and daily resource provisioning, executing autonomous real-time decisions without direct managerial supervision.

LEADERSHIP & CIVIC ENGAGEMENT

- **President, Spanish National Honor Society (Spring 2023 – Summer 2024):** Organized community outreach initiatives and bi-weekly assemblies, leveraging bilingual skills to bridge English- and Spanish-speaking community groups.
- **Webmaster, Cal Poly Humboldt Aviation Club (Fall 2024 – Present):** Built and optimized the club's web platform to support member recruitment and community airport events.
- **Sustained Community Volunteer (Spring 2023 – Present):** Contributed 100+ volunteer hours maintaining trails and facilities across Veterans' Hall, Volunteer Trail Steward, and Trinidad Coastal Land Trust.